Effectiveness of Deep Learning Technologies to Assist Educational Inclusion for Children with Down Syndrome

Shehan Silva
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
shehansilvak@gmail.com

Anjula Jayasinghe
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
anjulajay@gmail.com

Rishitha Dilshan
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
rishithadsh@gmail.com

Chandur Dissanayake
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
chandur.work@gmail.com

Samitha Vidhanaarachchi
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
samithapva@gmail.com

Dilshan De Silva
Department of Computer Science and Software Engineering
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
dilshan.i@sliit.lk

2023 5th International Conference on Advancements in Computing (ICAC) | 979-8-3503-5813-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICAC60630.2023.10417312

***Abstract*— Down Syndrome (DS) is a genetic disorder causing intellectual disability and developmental delays. Despite instances of discrimination, several individuals with DS have achieved success through proper education and community support. This research addresses the challenges faced by children with DS in terms of motor skill development, cognitive impairments, and speech difficulties with the use of Deep Learning Technologies such as LSTM, CNN machine learning models. To bridge the educational gap and improve speech intelligibility, this study focuses on oral motor exercises that enhance articulation skills. Cognitive development in individuals with DS is also a concern, prompting the proposal of a self-learning system to foster cognitive growth. Fine motor skills improvement is explored through yoga exercises, and interventions like interactive technologies are considered to enhance learning experiences. This paper discusses the development of a user-friendly mobile application for early intervention in DS, integrating interactive activities using advanced technologies and a parent therapy dashboard. The system's potential for improving developmental skills and future expansion is emphasized.**

Keywords—Down Syndrome, Inclusion, CNN, LSTM, YOLOV8, MediaPipe

I. INTRODUCTION

Down Syndrome (DS) is known as the most common genetic mental impairment that leads to intellectual disability. Usually, individuals with DS are neglected in education and society mainly due to communication difficulties caused by reduced speech and language intelligibility. This reduced speech and language intelligibility of individuals with DS is a result of physical and cognitive problems, including smaller voice cords, larger tongues, weaker muscles around the mouth, and poor memory and concentration [1,2]. The Down Syndrome Association has published a report detailing instances of discrimination, including denial of surgeries and demotivating actions or words related to the individual's condition [3].

Despite this discrimination, many individuals with DS have achieved success in life through proper education. For instance, Michael Beynon is an entrepreneur who sells his own product and Steven Brandon is an actor, among many others who have been able to lead a "normal life" with the support of an inclusive community and proper education [4]. Research has proven that self-paced computer-assisted instruction has been found to be beneficial for both students with DS and general education students [5]. The biggest challenges of children with DS is to overcome their difficulties in speech, cognitive skills , fine/gross motor skills which creates an educational gap from normal developing children thus making them fall behind in education. Researches conducted to date suggest that speech and language dysfunction and oral motor dysfunction are often associated together [6]. Children with DS frequently have lower muscle tone, their ability to move and coordinate their muscles in the mouth is impacted resulting in reduced oral motor skills [7].

Current studies show that oral motor exercises can help improve speech intelligibility and better articulation skills of children with DS [1], [8], [9]. These exercises aim to improve the strength, coordination, and control of the facial muscles important for speech such as the tongue, lips, and jaw. Children with DS often encounter a notable obstacle in the form of delayed cognitive development (CD) [10]. To address this issue, a proposed system has been devised, comprising elements such as self-learning, problem-solving, and engaging in mathematical activities. In recent years, various interventions and tools have been employed to help children with DS with their fine motor skills. One such approach is the use of yoga exercises, which have been shown to enhance muscle strength, coordination, and movement familiarity [11].

Children and adolescents with DS may experience delays in language development, with a weaker use of expressive language as they age. They would begin to talk later than typical children, also having difficulty in producing clear speech. [14,15]. Even though there are physical aspects that prevent them from normal speech development, researchers

979-8-3503-5813-1/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.

have found that this can be developed with repetition and practice [15]. Verbal short term memory, difficulty to store the precise phonological pattern of words are some understood reasons for this problem. Additionally, interactive technologies, such as mobile applications, have become increasingly popular in educational and therapeutic settings [12]. These platforms can provide engaging, personalized, and accessible solutions for children with Down syndrome, improving their motivation and overall learning experience [13].

Proposed solution is aimed at helping children with DS overcome their developmental challenges. It is designed as a mobile application platform serving the dual function of assistive technology and a developmental aid. The primary objectives encompass multiple facets, including the enhancement of speech , fine motor , gross motor, cognitive and oral motor skills. The ultimate aim is to bridge the educational gap between children with DS and typically developing children, fostering inclusivity in society.

II. Background Literature

Education can be particularly challenging for children with Down Syndrome (DS), primarily owing to difficulties they encounter in speech and language, reading and writing, as well as foundational mathematical skills. According to a study by Fidler and Nadel [16], early intervention programs that educate and assist kids with DS can greatly enhance their cognitive and verbal development. In this context, leveraging smart systems could serve as a potent tool for augmenting the educational support provided to children with DS. The approach presented by Fatemeh Javadi and colleagues [17] involves interactive games designed to facilitate speech development. The system integrates the MFCC1 algorithm to extract relevant audio cues from spoken language. Subsequently, the derived audio signals are subjected to comparative analysis against the original recordings using the DTW2 algorithm.

The JECRIPE game by André Brandão [18] is a system to enhance the cognitive and motor abilities of children with DS by means of a series of activities that involve mimicking designated movements. The movement recognition module of the game is responsible to recognize if the movements are correct or not. Notably, the said module has been designed to detect various body parts, including hands, head, and neck. To address this comprehensive recognition task, the authors have adopted a skin segmentation technique that operates in real-time and capitalizes on GPU processing.

In the work of V. Robles-Bykbaev [19], a system is introduced with the capability to formulate personalized educational intervention strategies. This system comprises two core algorithms: the profile analyzer and the intervention plans assembler. The profile analyzer plays a pivotal role in identifying the key attributes specific to each child, creating a structured data framework. This data structure is subsequently integrated with the curriculum materials through the intervention plans assembler. To optimize educational content selection, the system employs a rules-based reasoning module. Through an experiment involving 21 participants ranging from 6 to 11 years old, the researchers observed that most participants expressed satisfaction with the system's performance.

The author, D. Lima [20] designed a system with the aim of aiding individuals with DS in improving their literacy skills through a series of activities, all the while keeping track of their focus levels and identifying their facial expressions. The system comprises vowel matching, pairing vowel encounters, word pairing, and sound matching activities. Operating concurrently with these activities, a biofeedback algorithm known as the Attention Meter continually monitors the user's attention. The initial training of the biofeedback algorithm involved 80% of a dataset comprising 240 images and the algorithm demonstrated an overall success rate of 66%. The analysis has revealed that the biofeedback algorithm encountered difficulties primarily in distinguishing between the emotions of disgust and anger. However, after thorough examination of the gathered statistical data by the system, the researchers were able to successfully pinpoint the specific areas in which children encountered challenges. This analysis serves as compelling evidence for the system's efficacy.

The assistive learning platform, proposed by Sahan W. [21], encompasses a range of activities designed to enhance language, mathematics, social, and motor learning skills, as well as impart healthy habits to children with DS. The system employs a Deep Convolutional Neural Network (DCNN) model to discern shifts in a child's emotions during their involvement in activities. The objective of the model is to recommend activities according to the identified emotions, encompassing preferred quiz activities, music selections, and engaging videos. The model underwent training using a substantial dataset comprising labeled images, amounting to over thirty-five thousand samples. This trained model demonstrated the capacity to predict emotions across three distinct categories: happiness, sadness, and neutrality, achieving an impressive overall accuracy rate of 83%. However, variations in the quantity of available images for each individual class led to varying levels of precision in the classification process.

Most of the previous work related to this study has primarily focused on improving a single specific area of children with DS. Additionally, none of the previously discussed research has focused on the development of oral motor skills, which play a crucial role in speech-language intelligibility in children with DS. In contrast, the proposed solution aims to develop four key skills: speech-language, oral motor, cognitive, and fine motor skills by incorporating various fun and engaging activities within the application. Furthermore, the proposed solution has the capability to generate personalized plans for children and produce reports that can be utilized by parents and therapists, a feature lacking in most previous solutions. However, the activities included in their system consist of simple games that do not have the same focus on improving the skills in children with DS.

III. Methodology

Figure 1 depicts the system diagram detailing the envisioned solution's architecture, which aims to facilitate advancements in the aforementioned domains. Central to this approach is the utilization of deep learning models across critical developmental sectors. These models will be encapsulated within a Flask server, serving as the computational backbone. Concurrently, the orchestration of results will transpire through a NodeJS server, seamlessly integrated with a MongoDB database for efficient data management.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.

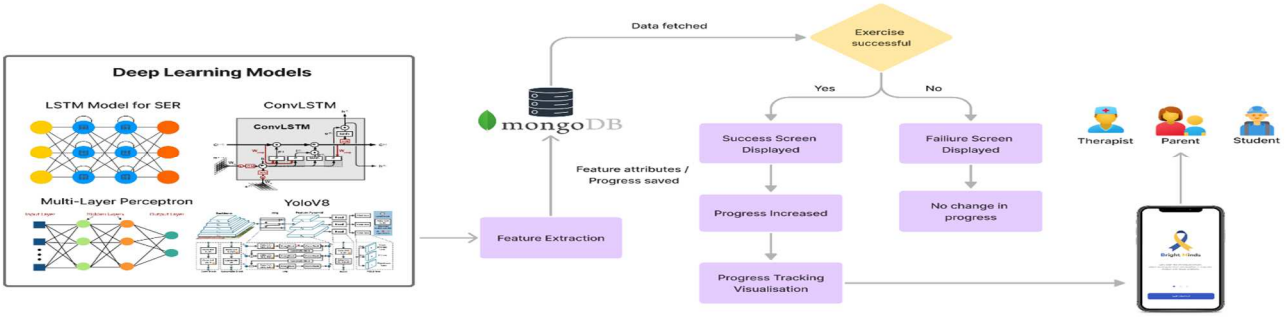


Fig. 1. System Diagram of the proposed solution

A.Data collection and preprocessing

A deep learning model was trained on a dataset of 107 videos to analyse the accuracy of oral motor exercises, encompassing Jaw Drop, Cheek Puff, and Pucker Face exercises. The videos underwent pre-processing, including frame selection, cropping, facial landmark marking, and resizing to 128x128 dimensions for uniformity. Additionally, a speech emotion recognition model was developed using 200 audio files for each of the 5 emotions (sad, angry, happy, frustrated, fear), sourced from an online dataset. For research on fine/gross motor activities, two methodologies - Angle Heuristics and Hand Gesture Recognition - were used, relying on the accurate mapping of key points to video frames using the MediaPipe platform. These models utilized a substantial dataset of 4450 key point records for hand gestures and 5296 records for finger gestures, which contributed to their robust gesture recognition performance. Furthermore, the COCO dataset was employed for self-exploration object detection activity as per Senehasa Children Resource Centre's guidelines. Total of 929 images were obtained to fine tune the model.

B. Speech Development - Vocabulary Activities

The solution consists of an activity which is targeted to leverage the spoken skills of pre-school children with Down Syndrome to help them bridge the gap for education. This will involve the student in identifying the word and speaking it out. The generated audio file is first sent to the node server from which it gets sent to the flask server. Inside the flask server, the correctness of the spoken word will be checked against the label, this is done using google cloud speech-to-text. First, the audio will be converted to text and the resulting text will be compared to the label using the 'Levenshtein algorithm' which outputs a similarity score. This similarity score will both be used to decide correctness and return the appropriate feedback. Additionally, this system will also determine the emotion of the children using a speech emotion recognition model, this also will determine the feedback to be returned.

The speech emotion recognition model has been created using LSTM (Long Short-Term Memory) neural networks designed to automatically detect and classify emotions from spoken language. LSTM models are a type of recurrent neural network (RNN) known for their ability to capture long-term dependencies in sequential data. In the context of speech emotion recognition, this LSTM model is trained on a dataset containing labeled speech samples associated with specific emotions. The model learns to extract relevant features from the acoustic properties pitch, energy, Mel-frequency cepstral coefficients (MFCCs), and prosodic features like duration and intonation, to predict the emotional state conveyed in the utterances. By utilizing the memory cells within LSTM networks, the model can effectively capture temporal dependencies and nuances in the speech signal, enabling accurate emotion recognition even in longer speech sequences.

The model is a sequential model, meaning the layers are stacked in a linear sequence. It starts with an LSTM layer that has 123 units. The LSTM layer is designed to process sequential data and capture temporal dependencies. In this case, the input sequences are expected to have a length of 40-time steps and a single feature. The LSTM layer is followed by a dense layer with 64 units, which applies a rectified linear activation function (ReLU) to introduce non-linearity in the model. A dropout layer with a dropout rate of 0.2 is added after the dense layer to help prevent overfitting. Another dense layer with 32 units and a ReLU activation follows, further extracting relevant features. Another dropout layer with a dropout rate of 0.2 is added for regularization. The final layer is a dense layer with 7 units and a SoftMax activation function, which outputs probabilities for the 7 emotion categories. The model is compiled with the categorical cross-entropy loss function, which is suitable for multi-class classification, and the Adam optimizer is used for optimization. Overall, this model architecture leverages LSTM and dense layers to extract features and make predictions on sequential data, specifically aimed at predicting emotions from the provided audio input.

C. Speech Development - Oral Motor Activities

The current solution can analyse the movements of the facial muscles, including the tongue, lips, and cheeks, to assess the child's performance during each exercise. Points will be rewarded to the child based on performance, considering the child's age, the accuracy of the child's muscle movements, and the level of difficulty.

The accuracy analysis of facial muscle movements during oral motor exercises is a crucial component of the system, as it drives the reward system, exercise plan generation, and progress tracking for children. To achieve this, a Convolutional LSTM (ConvLSTM) model, which combines Convolutional Neural Networks (CNN) and Long-Short Term Memory (LSTM), is utilized. The ConvLSTM model has proven effective in video classification tasks and has been trained using the collected videos of individuals performing oral motor exercises. At present, the model focuses on predicting the accuracy of three specific oral motor exercises, Jaw Drop, Cheek Puff and Pucker Face. A video recording of the child performing the exercise will be used as the input to make the prediction using this model.

The ConvLSTM model comprises five key layers. The initial ConvLSTM2D layer captures spatial and temporal

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.

features by analysing neighboring frames with a 2D Convolutional LSTM using 4 filters, (3,3) kernel, and hyperbolic tangent activation. Subsequently, 3D MaxPooling retains vital features while reducing spatial dimensions. To curb overfitting, Time Distributed Layer applies dropout to each time step's output. The Flatten Layer transforms this for compatibility with dense layers, extracting high-level features. Lastly, the Dense Layer, using SoftMax, generates class probabilities. This configuration enables comprehensive exercise video analysis.

The training process was carried out over 50 epochs. To ensure variability in the training process and avoid any potential bias, the data was shuffled prior to each epoch. To assess the model's performance during training, a portion of the training data (20%) was set aside as a validation set. The validation split helps evaluate the model's generalization ability and detect overfitting. The model's performance on the validation set was monitored throughout the training process. The rewarding system plays a significant role in the oral motor exercises component of the system, where the child receives points based on their performance. This point allocation takes into account the child's age, and the exercise accuracy, with accuracy being predicted by the ConvLSTM model. The system utilizes a decision tree model for determining the number of points to reward. To train the model, a dataset was created with three columns representing the features and labels: accuracy, age, and points. The dataset was organized according to different accuracy, age, and points values. The features and labels were then split into two arrays, 'X' and 'y', where 'X' contains the feature values (accuracy, age) and 'y' contains the corresponding label values (points).

D. Cognitive Skills Development

One of the primary goals of the proposed smart system is to augment and optimize a child's cognitive abilities, encompassing essential facets such as critical thinking, efficient problem-solving, and enhanced memory skills. To facilitate interactive learning of new objects, a self-exploration object identification activity is designed using the coco dataset. This activity involves capturing an image and predicting the object using a single-stage object detection model, YOLOv8 (You Only Look Once Version 8). The model utilizes the Darknet-53 backbone network, which offers improved speed and precision compared to the previous YOLOv7 network [22] network. To achieve a balance between prediction speed and optimal accuracy, the proposed approach employs YOLOv8m for the specific activity. While YOLOv8n exhibits faster prediction results, YOLOv8m is specifically selected to prioritize attaining the utmost accuracy levels. By leveraging YOLOv8m, the system ensures reliable and precise object identification, thereby facilitating enhanced performance in the given context.

The obtained results encompass essential information such as class labels, confidence scores, and normalized coordinates of the identified bounding boxes. These values are then employed to construct prediction objects, encapsulating pertinent details such as the class ID, class name, confidence score, and bounding box coordinates. These prediction objects are subsequently stored in designated collections, enabling comprehensive analysis and further processing.

To enhance a child's mathematical skills, a counting exercise is provided where the child demonstrates their answers using their fingers. The accurate detection of hand landmarks is of utmost importance in this exercise, and that's where the MediaPipe library comes into play. By leveraging the capabilities of the MediaPipe library, precise hand landmark detection is achieved. This functionality enables intuitive and natural user interfaces, allowing children to interact with digital content in a more immersive and instinctive manner. The MediaPipe library ensures the accurate positioning of the thumb by comparing the x-coordinates of the thumb tip and metacarpophalangeal (MCP) landmarks. Based on the hand label and the outcome of this comparison, the library updates the thumb status and increments the thumb count for the corresponding hand. This functionality is crucial for correctly identifying the hand gesture and ensuring accurate counting during the exercise.

E. Motor Skills Development

Motor skills development is another area which is aimed to be developed also under a technology-centric approach. A key aspect of this methodology is the computation of an individual's progress as a percentage, calculated by assessing the successful completion of tasks in conjunction with the subsequent processing and execution. A significant component of this research methodology is the calculation of a final score, accumulated through successful completion of tasks. This score is instrumental in generating a weekly progress report, which is then shared with key stakeholders for longitudinal tracking of developmental progress.

The MediaPipe pipeline initiates with a sophisticated person detection algorithm, designed to identify and isolate the subject within an image or video frame. This was succeeded by a landmark mapping operation that intricately assigns landmarks onto the detected body, a methodological step analogous to the Pose Landmark model. However, in the case of Angle Heuristics, the focus pivots toward the computation of angles between the identified key points, which is a requisite for efficient action classification. The reliability and precision of MediaPipe can be attributed to its comprehensive training on an array of significant datasets, including COCO, MPII, and proprietary datasets from Google. Angle Heuristics. This methodology classifies a spectrum of poses by scrutinizing the angles formed between different body joints, thereby applying geometric principles for a nuanced understanding and categorization of body poses.

The Hand Gesture Recognition methodology, in contrast, adopts a Multi-Layer Perceptron (MLP) model.

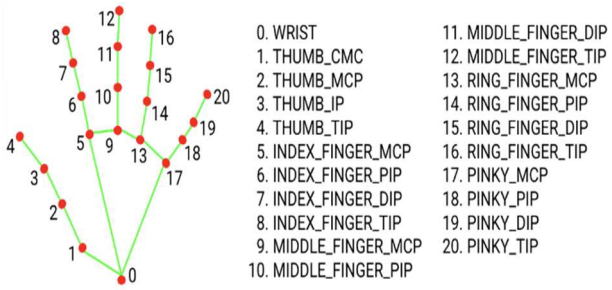


Fig. 2. Hand Key Point Mappings

This model utilizes the detected key points as input, thereby enabling the recognition of a variety of hand signs and finger gestures as illustrated the Fig. 2.To accommodate the

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.

existence of two distinct gesture types, our approach integrates two separate TensorFlow Lite models, each equipped with its corresponding trainable dataset to optimize recognition tasks.

The system architecture designated for hand gesture recognition is rooted in a dual neural network paradigm, comprising Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM) networks. MLPs demonstrate proficiency in modeling non-linear relationships, while LSTMs are adept at processing sequential data. Therefore, the system relies on MLPs for hand gestures, whereas for finger gestures, which inherently involve continuous movements, a blend of MLP and LSTM networks is deployed. In summary, the methodologies employed in this research encompass an advanced, interconnected system architecture that seeks to create an interactive, engaging, and personalized platform aimed at enhancing the required skills for children with Down Syndrome.

IV. Results and Discussion

A. *Analysis of speech development*

The accuracy analysis of the children's facial muscle movements during the exercises was conducted using the ConvLSTM model. Currently the model is capable of successfully predicting the accuracy of three different exercises with an overall accuracy rate of 92% as shown in the Fig.3. This indicates the effectiveness of the ConvLSTM model in assessing and analyzing the children's performance.

```
# Evaluate the trained model.
model_evaluation_history = convlstm_model.evaluate(features_test, labels_test)

1/1 [==============================] - 7s 7s/step - loss: 0.1470 - accuracy: 0.9259
```

Fig. 3. Performance of ConvLSTM model

As depicted in the Fig. 4. ConvLSTM model has been trained for 30 epochs. In the future, it is planned to further enhance the system by expanding the number of exercises and increasing the size of the dataset. By incorporating a wider range of exercises, a more comprehensive assessment of the children's performance across various oral motor tasks can be provided. Additionally, a larger dataset will enable the ConvLSTM model to learn from a broader range of examples, enhancing its accuracy and generalizability.

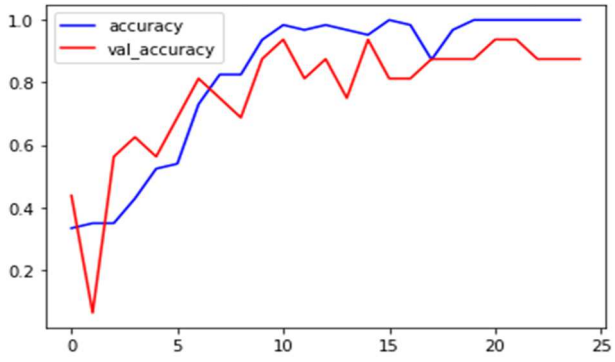


Fig. 4. Total Accuracy vs Validation Accuracy - ConvLSTM model

The decision tree model performed remarkably well in predicting the appropriate point allocations. The model was successful in determining the ideal number of points to award by careful examination of the child's age, and exercise accuracy. This personalized approach ensures that each child received a fair and tailored reward based on their individual abilities and progress. Furthermore, the reward system's effectiveness extends beyond immediate engagement, as it also encourages the children to persist in their efforts and strive for continuous improvement. The ability to unlock higher difficulty levels through successful completion of exercises provides a sense of achievement and encourages the children to push their limits and enhance their speech and language skills.

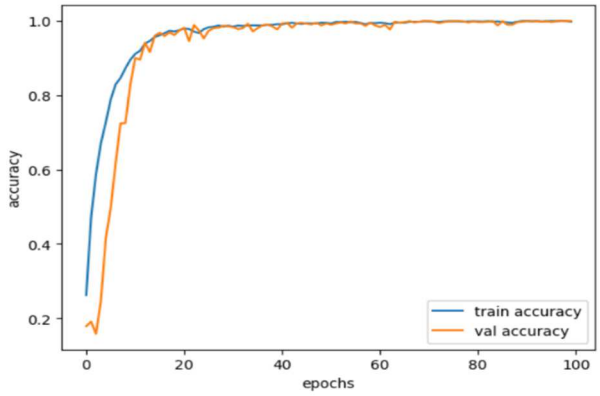


Fig. 5. Training Accuracy vs Validation Accuracy – Sentiment Identification model

Figure 5. illustrates the accuracy measurement for the sentiment identification model used for the vocabulary development activity which has been trained for 100 epochs, the overall accuracy level reaches 99.75% which can be considered as a highly accurate model. Validation Accuracy and Train accuracy are increasing and nearly equalize to each other, which means there is less overfitting nor underfitting. The identification of the sentiment has been impactful in deciding the best activities for the child and to identify the aspects for their relevant development. Ccustomized activities are tailored from the therapist also considering the mental state of the child as a valuable insight.

B. *Analysis of Cognitive Development*

The COCO dataset, a large collection of images for object detection tasks, was utilized to improve the training process and obtain valuable insights of self-exploration object identification activity. The YOLOv8m model architecture, which is known for its efficiency and accuracy in detecting objects, was used for the training phase. The model was trained for a total of 50 epochs, which ensures that it learns from the dataset multiple times to improve its performance. To ensure compatibility and computational efficiency, all images were resized to 640 dimensions. During the object detection process, confidence scores were obtained for each identified object in the images. However, only the maximum confidence score for each detected object was kept instead of keeping all the confidence scores. This approach helps to prioritize the most accurate and reliable detection. This approach, which uses the COCO dataset and YOLOv8m model, is expected to facilitate effective training and obtain valuable insights from the object identification process. This will ultimately lead to improved performance and accuracy in object detection tasks.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.

C. *Analysis of motor exercises*

The fine motor exercise recognition model achieved an overall accuracy of 0.89, as shown in the Fig.6. In essence, the model correctly identified 89% of the gestures in the test set. The macro average row represents the average precision, recall, and F1-score across all classes without taking class imbalance into consideration. The weighted average row, on the other hand, provides the average precision, recall, and F1-score but considers the class imbalance.

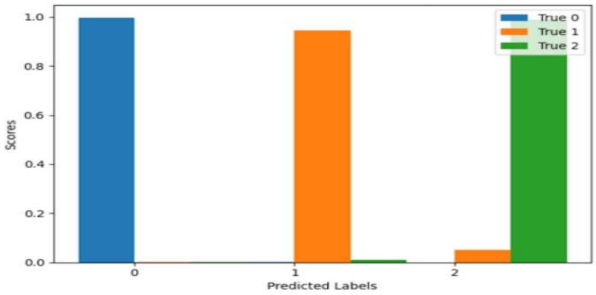


Fig. 6. Fine Motor Skills Model Accuracy

The macro average precision, recall, and F1-score are all 0.89, which is consistent with the overall accuracy. This suggests that the model is effective in classifying gestures based on the input data, regardless of the class imbalance. The model also achieved strong precision, recall, and F1 scores for each individual class. For example, the precision, recall, and F1 scores for the "circle" class are 0.90, 0.91, and 0.90, respectively. This suggests that the model is particularly good at identifying the "circle" gesture. Overall, the fine motor exercise recognition model achieved a high level of accuracy and precision. This suggests that the model is a promising tool for classifying fine motor exercises.

V. CONCLUSION

Children with Down syndrome are facing different developmental challenges such as motor skill development, cognitive impairments, and speech difficulties. We have proposed a user-friendly mobile application that can be used for early intervention for children with Down syndrome. The application includes interactive activities that are designed to capture the attention of users. The activities were implemented using a variety of technologies, including convolutional neural networks, decision tree algorithms, and image processing. CNN, YOLOV8, and ConvLSTM models with accuracies between 92% and 99% were obtained to improve the performance of exercises. Furthermore, the system includes a parent therapy dashboard to provide more assistance. In the future, the system will be expanded to include more interactive games that target a wider range of developmental skills. This will allow users to have fun while learning in an optimal way.

REFERENCES

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] G. Laws and D. Bishop, "Pragmatic language impairment and social deficits in Williams syndrome: a comparison with Down's syndrome and specific language impairment.," International journal of language & communication disorders, vol. 39, no. 1, pp. 45-64, 2004.

[3] “Intellectual Disability Info”, [Online]. Available: http://www.intellectualdisability.info/family/articles/parents-views [Accessed: 07-Feb-2023].

[4] “Famous and Successful people with downs syndrome”, [Online]. Available: https://disabilityhorizons.com/2022/10/10- famous-and-successful-people-with-downs-syndrome/ [Accessed: 07-Feb-2023].

[5] G. Wolpert, “What general educators have to say about successfully including students with Down Syndrome in their classes,” Journal of Research in Childhood Education, vol. 16, no. 1, pp. 28–38, 2001. doi:10.1080/02568540109594972

[6] K. Alcock, "The development of oral motor control and language.," Down's syndrome, research and practice : the journal of the Sarah Duffen Centre, vol. 11, no. 1, pp. 1-8, 2006

[7] B. Dodd, “Differential diagnosis of pediatric speech sound disorder,” Current Developmental Disorders Reports, vol. 1, no. 3, pp. 189–196, 2014. doi:10.1007/s40474-014-0017-3

[8] C. Stoel-Gammon and J. A. Cooper, "Patterns of development in phonology and morphology in children with Down syndrome.," Journal of speech and hearing research, vol. 27, no. 3, pp. 429-438, 1984.

[9] C. Stoel-Gammon and K. Otomo, "Babbling and first words in children with," Applied psycholinguistics, vol. 7, no. 3, pp. 229-241, 1986.

[10] S. Udbhasa, P. Lelkada, M. Priyanka, E. Senarathna, S. P. Vidhanaarachchi, J. Wickramatathne, and Janaka L. Wijekoon, "Interactive, Visual-Learning based Tool for Hearing Impaired Children to Improve Language and Cognitive Skills," 2023 International Conference on Information Technology (ICIT), Amman, Jordan, 2023, pp. 416-421, doi: 10.1109/ICIT58056.2023.10225863.

[11] M. L. Galantino, R. Galbavy, and L. Quinn, “Therapeutic Effects of Yoga for Children: A Systematic Review of the Literature,” Pediatric Physical Therapy, vol. 20, no. 1, pp. 66–80, 2008.

[12] S. Johnson, D. Lamb, L.Marston, D. Osborn, O. Mason, C. Henderson, G. Ambler, A. Milton, M. Davidson, M. Christoforou, S. Sullivan, and R. Rachael, “Peer-supported self-management for people discharged from a mental health crisis team: a randomised controlled trial,” The Lancet, vol. 392, no. 10145, pp. 409–418, Aug. 2018.

[13] S. Boucenna, A. Narzisi, E. Tilmont, F. Muratori, G. Pioggia, D. Cohen, and M. Chetouani “Interactive Technologies for Autistic Children: A Review,” Cognitive Computation, vol. 6, no. 4, pp. 722–740, May 2014.

[14] J. E. Roberts, J. Price, and C. Malkin, “Language and communication development in Down Syndrome,” Mental Retardation and Developmental Disabilities Research Reviews, vol. 13, no. 1, pp. 26–35, 2007. doi:10.1002/mrdd.20136

[15] D. Dodd, P. McCormack, and G. Woodyatt, "Evaluation of an intervention program: Relation between children's phonology and parents' communicative behavior," American Journal on Mental Retardation, vol. 98, no. 5, pp. 632–645, 1994.

[16] D. J. Fidler, and L. Nadel, "Education and children with Down syndrome: Neuroscience, development, and intervention," Mental retardation and developmental disabilities research reviews, vol. 13, no. 3, pp. 262-71, 2007.

[17] U. F. Javadi, F. Fotouhi Ghazvini, and R. Dianat, "Mobile Speech Therapy Application Using Speech Processing for Intellectually Disabled," Journal of Medical Signals and Sensors, vol. 11, no. 4, pp. 209-218, 2018.

[18] A. Brandão, E. Passos, C. Vasconcelos, A. Conci, E. Clua, P. T. Mourão, and M. Cordeiro, “Stimulating imitation of children with Down syndrome using a game approach,” VIII Brazilian Symp. Games Digit. Entertain., pp. 97–100, 2009.

[19] V. Robles-Bykbaev, E. Andrade-Prieto, P. Solórzano-Guerrero, Y. Robles-Bykbaev, F. Pesántez-Avilés, and A. Parra-Astudillo, "An educational support tool based on robotic assistants, mobile apps, and expert systems for children with Down syndrome," 2018 IEEE XXV International Conference on Electronics, Electrical Engineering and Computing (INTERCON), pp. 1-4, 2018.

[20] D. Lima et al., “Software with biofeedback to assist people with down syndrome,” International Journal of Computer Applications, vol. 158, no. 5, pp. 31–37, 2017. doi:10.5120/ijca2017912847

[21] S. Wellala, S. A. Thathsarani, D. Senaratne, P. Samaranayake and A. Jayakody, "Assistive Learning Platform for Children with Down Syndrome," 2020 20th International Conference on Advances in ICT for Emerging Regions (ICTer), Colombo, Sri Lanka, 2020, pp. 294-295, doi: 10.1109/ICTer51097.2020.9325487.

[22] C. Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, “Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors,” 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023. doi:10.1109/cvpr52729.2023.00721

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:46:36 UTC from IEEE Xplore. Restrictions apply.